

### Lab session:

Generate random numbers and check if they are in normal distribution using scipy libraries.

#### # Import Required Libraries

#### # Generate Random Numbers From Normal Distribution

```
data = np.random.normal(loc=50, scale=10, size=1000)

print(data[:10])
```

(loc = mean of distribution, scale = standard deviation, Size = no. of observations)

#### # Visualize the Distribution

```
sns.histplot(data, kde=True)
plt.title("Histogram of Generated Data")
plt.show()
```

```
import matplotlib.pyplot as plt

plt.hist(data)
plt.show()
```

#### #Using Shapiro test

```
# Perform Shapiro test
stat, p_value = stats.shapiro(data)
```

W close to 1: data is normal

p-value > 0.05: data is normal

W far from 1: data may not be normal.

P-value  $\leq$  0.05: data is not normal

#### # Using Pearson test

```
# Perform D'Agostino-Pearson Test
stat, p = stats.normaltest(data)
```

stat (small value): data is close to normal

p-value > 0.05: data is normal

stat (large value): Dat is far from normal

P-value  $\leq$  0.05: data is not normal

**Ques:** Load Iris dataset and find the following:

```
<> Python  
  
from scipy.stats import binom  
  
n = 10  
p = 0.5  
  
prob = binom.pmf(6, n, p)  
print(prob)
```

If 35% of flowers are Setosa, and randomly 10 flowers are picked from the dataset, find:

1. What is the probability exactly 4 flowers are Setosa?
2. What is the probability at least 7 flowers are Setosa?
3. What is the probability no flower is Setosa?
4. If petal length < 2 cm is considered “success,” what is the probability 6 out of 12 flowers satisfy this?

**Ques: Load Coffee Shop Dataset**

Using your coffee shop sales dataset:

Suppose the average number of customers entering the shop per hour is:

$$\lambda = 18$$

**Questions**

1. What is the probability exactly 20 customers enter in one hour?
2. What is the probability fewer than 10 customers enter?
3. What is the probability more than 25 customers enter?
4. What is the probability no customer enters between 3 AM and 4 AM?
5. What is the probability exactly 5 cold coffee orders occur in one hour if average cold coffee orders are 7 per hour?